

Literature status: determinants of random unitary pencils

Run `fa_banach_001` — lane 12

11 August 2026

Status: `literature_already_answered`.

Source question

Jury and Roman’s *Determinants of Random Unitary Pencils* [1] asks Conjecture (1.9) on PDF page 3. For a tuple $\mathcal{X} = (X_1, \dots, X_g)$ of $k \times k$ matrices, put

$$L_{\mathcal{X}}(\mathcal{U}) = I_k \otimes I_d + \sum_{j=1}^g X_j \otimes U_j, \quad \mathbf{rad}(\mathcal{X})^2 = \rho \left(\sum_{j=1}^g X_j \otimes \overline{X_j} \right).$$

The conjecture states that if $\mathbf{rad}(\mathcal{X}), \mathbf{rad}(\mathcal{Y}) < 1$, then

$$\lim_{d \rightarrow \infty} \int_{U(d)^g} \det L_{\mathcal{X}}(\mathcal{U}) \overline{\det L_{\mathcal{Y}}(\mathcal{U})} d\mathcal{U} = \det \left(I_k \otimes I_{k'} - \sum_{j=1}^g X_j \otimes \overline{Y_j} \right)^{-1}. \quad (1)$$

The same conjecture is restated at the start of Section 3 of the source.

Identification of the answer

Jury, van Rensburg, and Roman’s later paper *Free versions of the strong Szegő limit theorem* [2] explicitly identifies and settles the question. The sentence immediately preceding Theorem 1.6 on PDF page 4 says that the theorem “fully proves the conjecture left open” in [1]. The theorem assumes

$$\rho \left(\sum_{j=1}^g A_j \otimes \overline{A_j} \right) < 1, \quad \rho \left(\sum_{j=1}^g B_j \otimes \overline{B_j} \right) < 1,$$

and concludes precisely the limit in (1), with (A, B, N, ℓ) in place of $(\mathcal{X}, \mathcal{Y}, d, k')$. These are exactly the source’s outer-spectral-radius hypotheses after squaring.

The later paper proves more: Theorem 6.2 on PDF page 25 treats quotients of determinants of monic pencils, and explicitly notes that Theorem 1.6 is the case $C = D = 0$.

Scope and provenance

The match is exact, not merely a reformulation inferred by this run. The supporting authors include both source authors, cite the source, name its open conjecture, use the same matrix tuples and spectral-radius domain, and prove the same determinant limit. Thus no part of Conjecture (1.9) remains open in its stated scope. This note records literature status only and makes no claim of new mathematics.

The check used exact-title, arXiv-id, author, “random unitary pencil,” and “outer spectral radius” searches, followed by direct inspection of the two papers. Local copies of both PDFs accompany this note.

References

- [1] M. T. Jury and G. Roman, *Determinants of Random Unitary Pencils*, arXiv:2506.04400v1 (2025); J. Funct. Anal. 291 (2026), no. 6, 111555.
- [2] M. T. Jury, L. J. van Rensburg, and G. Roman, *Free versions of the strong Szegő limit theorem*, arXiv:2607.25980v1 (2026).